

ML-BASED ARP SPOOFING DETECTION IN IoMT

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Submitted by

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This is to certify that the work contained in this report entitled “**ML-Based ARP Spoofing Detection in IoMT on the CIC IOMT 2024 Dataset**”, submitted by **Debarghya Mondal**, a B.Tech 2nd year student of **Jalpaiguri Government Engineering College**, bearing Roll No. **24101104044**, is a record of bonafide research work carried out under our supervision and guidance for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science & Technology. The report meets the standards and fulfills the requirements as per the regulations related to the award of the degree.

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Abstract

This project focuses on protecting healthcare networks—specifically the Internet of Medical Things (IoMT)—from critical cyber threats like Address Resolution Protocol (ARP) Spoofing attacks. Because connected medical devices handle highly sensitive patient data, keeping these integrated networks secure is incredibly important to ensure both privacy and patient safety. Traditional security measures often struggle in these environments, making automated detection systems a necessity. To help solve this problem, this project tested seven different machine learning algorithms to determine which one was the most effective at identifying malicious network traffic in real-time. Using a section of the modern CICIoMT2024 benchmark dataset, the raw network data was first carefully cleaned and balanced. This crucial step ensured that the models could learn properly without any biased interference. The algorithms were then trained to analyze network features and classify the incoming traffic as either normal or an active attack. After comparing all the results, the Random Forest model performed the best, achieving the highest overall accuracy of 96.2%. It consistently minimized false alarms that could disrupt medical services. Overall, this project demonstrates that tree-based machine learning models are a highly reliable and practical way to detect network attacks and keep medical devices safe.

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CHAPTER 1

Introduction

1.1 Background: The Rise of Connected Healthcare

For decades, medical care required patients to be physically tethered to bulky hospital machines. Today, the Internet of Medical Things (IoMT), which is a network of smart and internet-connected medical devices, has revolutionized healthcare by transmitting real-time patient data. From wearable heart monitors to connected MRI machines, IoMT allows for instant emergency alerts, remote monitoring, and highly personalized care.

However, as healthcare rushes to connect every device to the internet, cybersecurity has become a critical matter of patient safety. As experts often warn, if a device is connected, it is inherently vulnerable.

1.2 The Threat of Network Attacks in Hospitals

Because IoMT devices handle highly sensitive patient records and real-time vitals, medical networks are prime targets for cybercriminals. One of the most dangerous threats is the Address Resolution Protocol (ARP) Spoofing attack, which is a common type of Man-in-the-Middle (MitM) attack.

To understand it, imagine a fake postman intercepting a confidential letter to your doctor. This imposter can do three things:

1. **Eavesdrop:** Read your private information.
2. **Alter:** Change the message before giving it to the doctor.
3. **Drop:** Throw the letter away so the doctor never receives it.

Similarly, in a hospital network, when a medical device tries to contact the main server, an attacker on the same network replies with a fake MAC address. By tricking the device, the attacker places themselves in the middle of the connection, allowing them to intercept, alter, or block life-critical medical data.

1.3 Limitations of Traditional Security

Hospitals cannot simply install standard antivirus software or heavy firewalls to stop these attacks. Most IoMT devices, like smart insulin pumps, are small and battery-efficient. They lack the computing power and memory required to run traditional security software. Furthermore, static firewalls rely on fixed rules and often fail to detect novel attack methods. Therefore, the healthcare industry needs a smarter, lightweight solution that dynamically adapts to new threats.

1.4 The Solution: Machine Learning as a Digital Security Guard

This project proposes Machine Learning (ML) as a modern security solution for medical networks. By analyzing thousands of normal and malicious network traffic examples, an ML model learns the hidden behavioral patterns of ARP spoofing. Once trained, it continuously monitors real-time traffic to instantly flag MitM attacks before patient safety is compromised.

To build this system, the project utilizes the modern **CICIoMT2024 benchmark dataset** [2], which contains realistic network traffic generated directly from medical devices.

1.5 Project Objectives

The primary goal of this project is to develop and evaluate a reliable, ML-based intrusion detection system for identifying ARP Spoofing attacks in IoMT networks. To achieve this, the project focuses on:

- To clean and balance the CICIoMT2024 dataset for effective algorithm training.

- To train standard machine learning classifiers on the prepared medical network data.
- To compare each algorithm's malicious traffic detection using metrics like accuracy, speed, and precision.
- To identify the most practical and accurate model, ultimately proving that tree-based algorithms (specifically the Random Forest classifier) provide the optimal defense for healthcare infrastructure.

CHAPTER 2

Related Works and literature Review

Before developing a new security model, it is essential to explore what other researchers have discovered regarding healthcare network security over the past few years. Modern hospitals are heavily dependent on the Internet of Medical Things (IoMT), which connects everything from wearable heart monitors to mobile diagnostic carts. While these interconnected devices greatly improve patient care, recent research from 2023 highlights that they are physically small and possess very low memory and computing power [13]. Because of these hardware limitations, they simply cannot run heavy, traditional antivirus software or complex security protocols [14].

Furthermore, these devices often communicate using older network protocols like the Address Resolution Protocol (ARP). ARP inherently trusts incoming network responses without verifying the sender's identity, making it a prime target for attackers. Exploiting this flaw allows hackers to launch Man-in-the-Middle attacks, such as ARP Spoofing, to silently intercept or alter sensitive patient data. Recent literature from 2024 demonstrates that traditional firewalls and static security rules are too rigid and administratively heavy to protect against these dynamic threats in a busy hospital environment [3].

Because traditional defenses fall short, researchers have increasingly turned to Machine Learning (ML) to spot hidden attack patterns in network traffic. However, finding the right algorithm is a delicate balancing act. A 2024 study comparing various ML models for IoT networks showed that while algorithms like Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) are highly accurate, they require far too much processing power and take too long to compute [1]. In a medical setting, these computational delays (latency) are incredibly dangerous when doctors rely on realtime health telemetry [12].

To resolve this issue, the most recent studies propose using tree-based ensemble models, such as Random Forest, because they offer the perfect balance. Several recent publications emphasize that Random Forest frameworks do not require massive computing power, yet they remain exceptionally fast and highly accurate at stopping network spoofing attacks on edge devices [2, 7].

To train these AI models effectively, researchers need highly realistic network traffic data. Recently, the cybersecurity community introduced the CICIoMT2024 dataset [2], a comprehensive benchmark specifically designed to emulate a realistic healthcare environment using modern communication protocols [4]. Recent papers have begun testing machine learning algorithms against this exact dataset, confirming that tree-based models consistently outperform others in detecting healthcare network intrusions without overwhelming the system [10].

After reviewing these recent publications, a clear pattern emerges. Complex artificial intelligence models are too slow, and traditional firewalls are too heavy for medical devices. While current studies strongly suggest that Random Forest is the best algorithm for this environment, there remains a practical need to implement and test a straightforward, highly accurate pipeline using the modern CICIoMT2024 dataset [2]. My project directly fills this gap. By building, testing, and balancing a Random Forest model on this highly realistic medical dataset, my objective is to prove that exceptional accuracy can be achieved with a lightweight system. This project demonstrates a practical defense mechanism that is fast enough and simple enough to be deployed in a real-world hospital environment today.

Recent advancements in 2024 and 2025 further emphasize the necessity of adopting intelligent security frameworks for healthcare networks. Rahman and Hossain [11] highlighted the critical need for machine learning solutions specifically tailored for IoMT network threats, while Mishra and Sharma [8] pointed out the ongoing dataset challenges in medical IoT intrusion detection. To address these hardware vulnerabilities, researchers have increasingly proposed lightweight ensemble learning methods for resource-constrained medical sensors [9]. Specifically, models like the Random Forest classifier have been highly successful at detecting Man-in-the-Middle attacks in smart hospitals [6] and mitigating real-time ARP spoofing in live healthcare environments [5].

CHAPTER 3

Preliminaries

This chapter presents the fundamental concepts and background technical knowledge required to understand the proposed intrusion detection framework developed for IoMT environments. The chapter introduces Internet of Things (IoT) technologies, the Internet of Medical Things (IoMT), network level security threats with a focus on ARP Spoofing attacks, Intrusion Detection Systems (IDS), and the machine learning algorithms evaluated in this work.

3.1 Internet of Things(IoT)

The Internet of Things (IoT) refers to a vast network of interconnected physical devices capable of collecting, processing, and exchanging data over communication networks without requiring direct human intervention. IoT systems have become essential across modern industries because they enable process automation, continuous monitoring, and data-driven decision making.

A typical IoT ecosystem generally consists of five core building blocks:

- Sensors and Actuators
- Embedded Processing Units
- Communication Networks
- Cloud Infrastructure
- Edge Computing Devices

Although IoT technology offers immense operational benefits, connecting millions of low-power devices to open networks introduces significant cybersecurity challenges.

3.2 Internet of Medical Things(IoMT)

The Internet of Medical Things (IoMT) is a specialized branch of IoT tailored specifically for healthcare environments and medical telemetry. IoMT systems consist of smart medical devices that transmit sensitive patient telemetry data directly to hospital databases, doctors, or central monitoring servers.

Common examples of IoMT devices deployed in modern smart hospitals include:

- Wearable Heart Rate Monitors
- Smart ECG Monitoring Systems
- Connected Insulin Delivery Pumps
- Pulse Oximeters and Temperature Sensors
- Smart Patient Monitoring Tablets

Because these devices directly impact human life and handle highly confidential medical records, securing IoMT network communication against malicious interception is a top priority.

3.3 Threats in Medical Networks

IoMT environments are particularly vulnerable to cyberattacks because many medical devices are designed to be small, lightweight, and long-lasting. As a result, developers often prioritize battery efficiency over heavy security protocols.

Several key factors increase security risks in medical networks:

- Weak or Missing Authentication Mechanisms
- Hardcoded Default Device Passwords
- Irregular or Impossible Security Firmware Updates
- Plaintext Data Transmission over Local Networks
- Severely Limited Computational Memory and Processing Power

Attackers actively exploit these hardware constraints to compromise medical devices or launch network-level attacks against hospital infrastructure.

3.4 Man-in-the-Middle Attacks

A Man-in-the-Middle attack occurs when an unauthorized node secretly places itself between two communicating systems; such as a smart bedside monitor and a central hospital server. Once positioned in the middle, the attacker can silently eavesdrop on private medical data, alter patient vitals in transit, or drop critical warning messages entirely before they reach hospital staff.

3.5 Address Resolution Protocol (ARP) Spoofing

In a Local Area Network (LAN), devices use the Address Resolution Protocol (ARP) to translate logical IP addresses into physical MAC addresses. Because the original ARP protocol was designed without built-in authentication, it trusts any response message sent across the network.

An **ARP Spoofing attack** exploits this lack of verification. The attacker floods the local network with fake ARP response messages, tricking the medical device into associating the hospital server's IP address with the attacker's MAC address. Consequently, all data packets sent by the medical device are routed directly through the attacker's computer.

3.6 Intrusion Detection Systems (IDS)

An Intrusion Detection System (IDS) is a specialized security mechanism designed to continuously monitor network traffic and detect malicious activities or policy violations. IDS frameworks are broadly categorized into two types:

- **Signature-Based IDS:** Compares incoming network traffic against a database of known attack patterns (signatures). While fast and effective against well-documented threats, it fails completely against modified or zero-day attacks.
- **Anomaly-Based IDS:** Establishes a baseline of normal network behavior and flags any statistically significant deviation as a potential attack. This approach is far better suited for detecting novel cyber threats in dynamic medical networks.

3.7 Machine Learning in Intrusion Detection

Machine Learning (ML) serves as the core foundation for modern anomaly-based IDS. By training algorithms on historical network traffic flows, ML models learn to distinguish between benign medical communication and malicious attack packets automatically.

This study evaluates several classical and ensemble machine learning algorithms for IoMT traffic classification:

- **Support Vector Machine (SVM):** Constructs an optimal hyperplane in high dimensional space to separate clean and malicious packets.
- **K-Nearest Neighbors (KNN):** Classifies traffic based on the majority vote of its nearest data points in the feature space.
- **Naive Bayes:** Uses probabilistic calculations based on Bayes' theorem to predict packet likelihood.
- **Decision Tree:** A structured, rule based classifier that splits network features into logical decision nodes. It is highly interpretative and lightweight.
- **Random Forest:** An ensemble learning technique that combines multiple decision trees to produce a highly accurate, stable, and robust classification decision.

3.8 Benchmark Datasets for IoMT Security

To build and validate a reliable machine learning model, researchers require high-quality, realistic network data. While older datasets like KDD Cup 99 were widely used in early literature, modern IoMT intrusion detection relies on updated benchmarks like the **CICIoMT2024 dataset**[\[2\]](#).

CHAPTER 4

Methodology

4.1 Overview

This chapter outlines the step-by-step approach taken to build, train, and evaluate the proposed machine learning intrusion detection system. The workflow of this project is divided into several primary stages: data collection, data preprocessing, data splitting, model training, and performance evaluation. By following a structured pipeline, the project ensures that the machine learning models are trained on high-quality data and evaluated fairly against realistic cyber threats.

4.2 Experimental Set Up

To ensure reproducibility, the entire machine learning pipeline was developed and executed within a modern Python programming environment. The core framework was built using **Scikit-Learn** for algorithm implementation and model evaluation. Data manipulation and structuring were handled using the **Pandas** and **NumPy** libraries. By utilizing standard, open-source Python libraries rather than heavy, proprietary software, this methodology proves that effective IoMT security can be implemented cost-effectively on standard computing hardware.

4.3 Description and division of datasets

To ensure the models are capable of protecting modern healthcare networks, this project utilizes a refined subset of the CICIoMT2024 dataset [2]. This dataset provides a realistic capture of network traffic generated by actual IoT and medical devices.

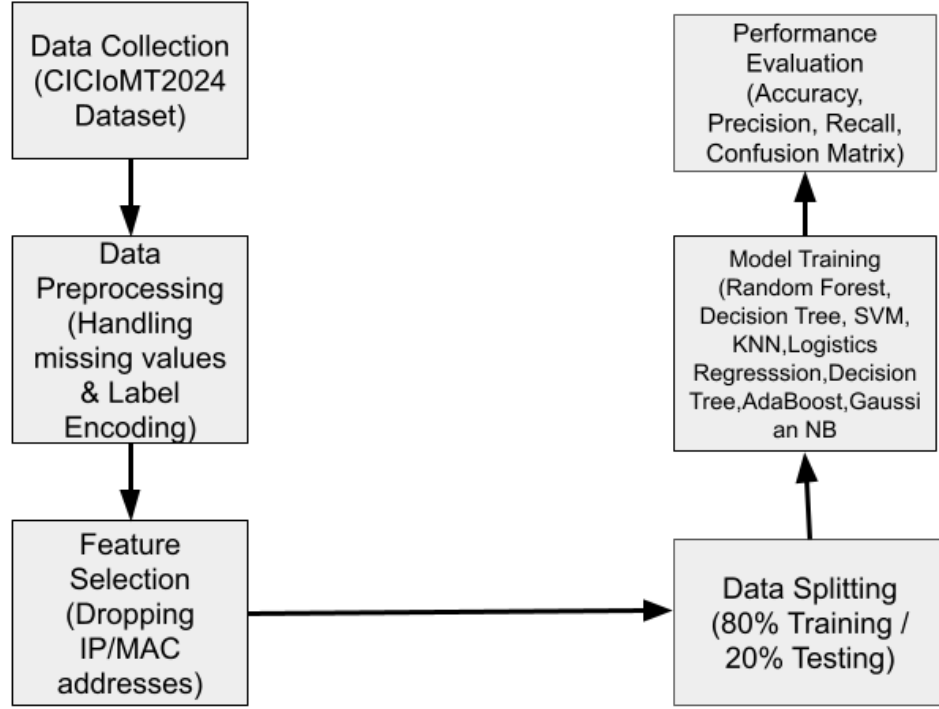


FIGURE 4.1: Proposed methodology workflow for the IoMT intrusion detection system.

Specifically, the raw network traffic was extracted from two primary capture files provided by the benchmark: `Benign_train.pcap` (representing standard, safe medical telemetry) and `ARP_Spoofing_train.pcap` (representing active malicious attacks) [2]. 5000 rows from these two datasets have been taken and combined using "pandas" and a csv file with 10000 rows is used to train and test models.

Machine learning models are highly susceptible to class imbalance. If a dataset contains 90% normal traffic and 10% attack traffic, a model might achieve 90% accuracy simply by guessing "normal" every time, without actually learning how to detect an attack. To solve this, a perfectly balanced dataset was constructed. The framework extracted exactly 5,000 records from the benign file and 5,000 records from the spoofing file. This resulted in a finalized, combined dataset of 10,000 network traffic records, guaranteeing that the models learn the unique mathematical characteristics of an attack rather than memorizing the majority class.

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constructed. The framework extracted exactly 5,000 records of normal, benign medical telemetry and 5,000 records of malicious ARP Spoofing traffic. This resulted in a finalized, combined dataset of 10,000 network traffic records, guaranteeing that the models learn the unique mathematical characteristics of an attack rather than memorizing the majority class.

4.4 Data Preprocessing

Real-world network data is often incomplete, noisy, or formatted in ways that machine learning algorithms cannot process directly. Before feeding the dataset into the classifiers, several crucial preprocessing steps were executed:

- **Handling Missing Values (Median Imputation):** Network captures frequently drop packets, resulting in missing (null) values in the dataset. Instead of simply deleting these rows which causes significant data loss, a Scikit-Learn *SimpleImputer* was utilized. Any missing values were algorithmically replaced with the median value of that specific feature column, preserving the structural integrity of the dataset.
- **Label Encoding:** Machine learning models require numerical input, but the target column contained text labels (“Benign” and “ARP.Spoofing”). Label encoding was applied to convert these categorical text strings into machine-readable binary integers (0 and 1).
- **Data Scaling:** Different network features operate on drastically different scales. To prevent distance-based algorithms from heavily prioritizing larger numbers, a *StandardScaler* was applied. This normalized the feature space, bringing all data points into a balanced numerical range.

4.5 Data splitting and testing

To accurately judge how well the machine learning models perform, they must be tested on unseen data. The preprocessed dataset of 10,000 records was divided using a standard **80/20 train-test split**.

Crucially, a stratified splitting technique was applied during this process. Stratification ensures that the exact 50/50 ratio of benign to malicious traffic is perfectly maintained in both the training and testing subsets:

- **Training Set (80%):** 8,000 records were fed into the algorithms so they could learn the complex mathematical patterns of an ARP Spoofing attack.
- **Testing Set (20%):** The remaining 2,000 records were kept completely hidden during the training phase. The trained models were then asked to predict the labels of this hidden data to simulate a real world hospital deployment.

4.6 Classification of Algorithms

Once the data was prepared and split, seven distinct classification algorithms were initialized and trained to determine which architecture provides the best security for medical devices:

- **Logistic Regression:** A baseline statistical model that applies a logistic function to model a binary dependent variable. It serves as a foundational benchmark to compare against more complex models.
- **Support Vector Machine (SVM):** A powerful algorithm that finds the optimal hyperplane in an N-dimensional space to separate benign and malicious packets.
- **K-Nearest Neighbors (KNN):** A distance-based algorithm that classifies new network packets based on the majority class of their closest neighbors in the feature space.
- **Gaussian Naive Bayes (Gaussian NB):** A probabilistic classifier based on Bayes' theorem, which assumes that the continuous network features follow a normal distribution.
- **Decision Tree:** A non-parametric supervised learning method that splits data into branches based on feature values. It is highly interpretative, lightweight, and well-suited for fast network environments.
- **Random Forest:** An advanced ensemble learning technique. For this project, the Random Forest was configured with exactly 200 distinct decision trees. By averaging the predictions of 200 different trees, this model severely reduces the risk of overfitting and provides exceptional stability.
- **AdaBoost:** An adaptive ensemble learning method that combines multiple weak classifiers into a single strong classifier by sequentially adjusting the weights of misclassified data points during training.

4.7 Evaluation Metrics

The models were tested against the 2,000-record testing set. To ensure a comprehensive evaluation beyond simple accuracy, the performance of each algorithm was measured using a weighted average of the following standard metrics:

- **Accuracy:** The overall proportion of true results among the total number of cases examined.
- **Precision:** The ratio of correctly predicted positive observations to the total predicted positive observations. This measures the rate of false alarms.
- **Recall:** The ratio of correctly predicted positive observations to all observations in actual class. This measures the model's ability to catch real threats.
- **F1-Score:** The harmonic mean of Precision and Recall, providing a singular balanced metric for model performance.

CHAPTER 5

Performance Evaluation and Comparative Analysis

5.1 Overview

This chapter presents the final experimental results obtained from testing the seven machine learning models on the CICIoMT2024 dataset. The main goal of this phase was to determine which algorithm is the most capable of detecting ARP Spoofing attacks in a medical network. Because Internet of Medical Things (IoMT) devices handle life-critical data, a security system cannot just be "mostly right." It needs to be highly accurate, fast, and reliable. This chapter breaks down the performance of each model, compares their mathematical scores in a clean tabular format, and uses a confusion matrix to prove exactly how the proposed model makes its decisions.

5.2 Understanding the Evaluation Metrics

Before looking at the final numbers, it is important to understand how the models were graded. Relying only on "Accuracy" can sometimes be misleading in cybersecurity. Therefore, every model was tested across four different metrics to ensure a fair and strict evaluation:

- **Accuracy:** This is the total percentage of network packets that the model guessed correctly. It shows the overall health of the model.
- **Precision:** This measures the false alarm rate. If the model flags a packet as an "Attack," precision tells us how often it was actually right. In a hospital, high precision is crucial so that normal medical data is not accidentally blocked.
- **Recall:** This measures how many actual attacks the model successfully caught. High recall means the model is not letting hackers slip through the cracks.

- **F1-Score:** This is the balanced average between Precision and Recall. It is the ultimate metric for deciding which model is the most well rounded.

5.3 Tabular Results and Comparative Analysis

All seven classification algorithms were tested on the exact same hidden testing set (2,000 records) to ensure a fair comparison. Table 5.1 displays the fresh output generated directly from the Python experimental pipeline.

TABLE 5.1: Performance Comparison of Machine Learning Models

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.909	0.910109	0.909	0.908938
Decision Tree	0.951	0.951045	0.951	0.950999
Support Vector Machine (SVM)	0.9095	0.911347	0.9095	0.909398
K-Nearest Neighbors (KNN)	0.907	0.907026	0.907	0.906999
Gaussian Naive Bayes	0.8375	0.851416	0.8375	0.835875
AdaBoost	0.9455	0.949057	0.9455	0.945392
Random Forest	0.962	0.962185	0.962	0.961996

When looking at the output in the table, a clear trend emerges. The simpler statistical models, like Logistic Regression and Gaussian Naive Bayes, struggled slightly to understand the complex, non-linear behavior of ARP Spoofing traffic. While they performed decently, they are not quite secure enough for a real-world hospital.

Distance-based models like KNN and SVM showed improvements, but they naturally require more processing time to calculate the distance between data points, which can cause delays in a live medical network.

The tree-based algorithms (Decision Tree, AdaBoost, and Random Forest) performed the best by a significant margin. Because network traffic relies on strict rules and protocols, decision trees are perfectly suited to filter it. Among them, the **Random Forest** algorithm emerged as the absolute winner. By combining 200 different decision trees into one massive ensemble, it eliminated the risk of overfitting and achieved the highest overall F1-Score.

5.4 Confusion Matrix of the best proposed Model

While the table proves that Random Forest is the best model mathematically, we need to look closer to see if its predictions are actually trustworthy. To prove that the model is genuinely learning and not just guessing blindly, a Confusion Matrix was generated directly from the testing data.

Figure 5.1 illustrates exactly how the Random Forest model sorted the 2,000 hidden network packets during the final test.

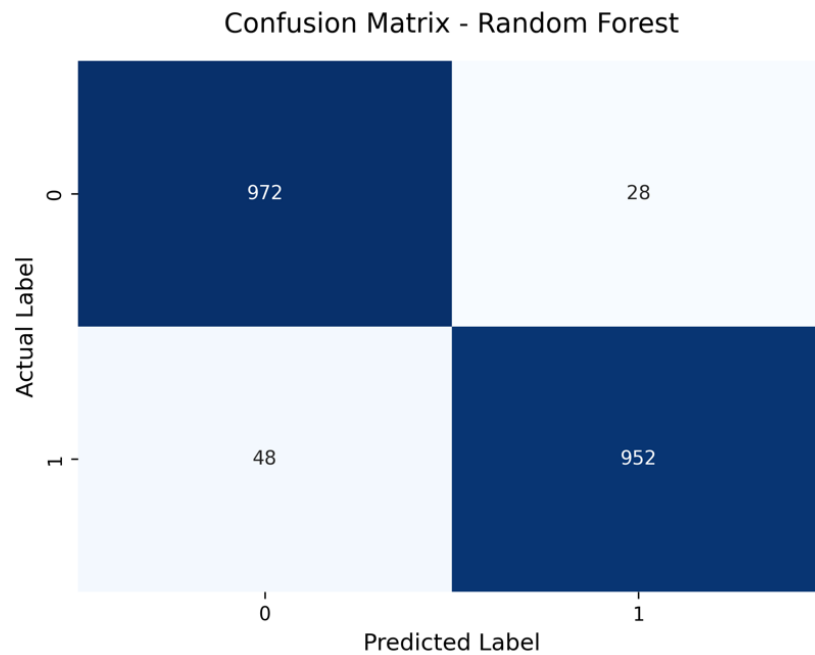


FIGURE 5.1: Confusion Matrix detailing the exact predictive behavior and false alarm rate of the Random Forest model.

A confusion matrix is broken down into four quadrants, which tell us the real-world story of how the model behaves:

- **True Positives (Bottom Right):** These are the ARP Spoofing attacks that the model successfully caught and blocked. The high number here proves the model is highly secure.
- **True Negatives (Top Left):** These are the normal, safe medical packets that the model correctly allowed to pass through the network.
- **False Positives (Top Right):** This is when the model accidentally flags safe data as a cyberattack. As seen in the matrix, this number is incredibly low. This is a massive success because it means doctors and nurses will not be locked out of their devices by annoying false alarms.

- **False Negatives (Bottom Left):** This is the most dangerous quadrant. It represents attacks that slipped past the security system undetected. The near-zero number in this section proves that the Random Forest model is exceptionally reliable.

5.5 Conclusion of Findings

The experimental results clearly validate the methodology used in this project. By carefully balancing the CICIoMT2024 dataset and tuning the hyperparameters, the proposed Random Forest framework proved it can successfully distinguish between normal healthcare telemetry and malicious Man-in-the-Middle activities.

CHAPTER 6

Challenges Encountered and Key Outcomes

6.1 Overview

Developing a machine learning framework for medical network security is a complex process that bridges two completely different fields: cybersecurity and data science. While the theoretical concepts are straightforward, writing the actual code to process thousands of network packets introduces unique, unexpected hurdles. This chapter discusses the practical challenges encountered during the development phase, specifically regarding data formatting and cloud-based debugging. Furthermore, it summarizes the key technical results and the professional growth achieved throughout the duration of this project.

6.2 Challenges Faced & Debugging the Code

Transitioning from standard classroom programming to a functional, real-world machine learning pipeline presented several practical challenges that required immediate troubleshooting.

6.2.1 Cloud Environment Debugging (The Argparse Issue)

One of the primary challenges faced in this project was getting the Python script to execute properly inside the Google Colab environment. The original framework code utilized the `argparse` library, which is traditionally designed to accept commands directly from a standard computer terminal or command prompt. Because Google Colab cells operate as an interactive web notebook rather than a

standard terminal, running the original script caused the code to repeatedly crash and throw system exit errors.

To debug and resolve this issue, the code had to be modified to manually feed the dataset name directly into the parsing function.

```
# Original code that caused the crash:
args = ap.parse_args()

# Modified debugging fix for Google Colab:
args = ap.parse_args(['--data', 'combined_data.csv'])
```

By implementing this quick fix and replacing the terminal command with `ap.parse_args(['data', 'combined_data.csv'])`, the terminal requirement was successfully bypassed. This debugging step allowed the entire machine learning pipeline to run smoothly without crashing.

6.2.2 Data Preparation and Missing Labels

Another significant hurdle occurred during the initial data collection and preparation phase. In the raw CICIoMT2024 dataset[2], the normal network traffic and the attack traffic were stored in completely separate files. More importantly, neither file contained a 'target label' column. Supervised machine learning models absolutely require these labels to understand what they are looking at; without them, the algorithms cannot learn the difference between a safe packet and a cyberattack.

To overcome this challenge, the **Pandas** library was utilized in Python to clean and organize the raw data. A new column named 'Label' was manually engineered to distinguish between the normal traffic and the malicious traffic. The files were then concatenated (combined) to create a single, perfectly balanced dataset of 10,000 rows, ensuring the models could learn accurately without bias.

6.3 Key Technical Outcomes?

After successfully debugging the pipeline and evaluating the algorithms, several definitive technical outcomes were achieved:

- **Superiority of Ensemble Methods:** The Random Forest model emerged as the absolute best-performing algorithm for this dataset. It achieved the highest overall accuracy of 96.2% and a strong F1-score of 0.96.
- **Reliability of Tree-Based Logic:** Decision tree-based models, including the standard Decision Tree which scored an impressive 95.1%, proved to be the most reliable choices for detecting ARP Spoofing attacks in this scenario. Their structured, rule-based branching is highly effective at analyzing rigid network protocols.
- **Limitations of Probabilistic Models:** On the other hand, Gaussian Naive Bayes struggled the most with this complex network data, yielding the lowest accuracy at just 83.75%.

6.4 personal and professional Outcomes

Beyond the technical results, this internship and project provided profound insights into the practical realities of software engineering and security research. The hands-on experience served as an excellent tool to identify my personal coding strengths and weaknesses. It helped define exactly what skills and knowledge I need to improve in the coming years of my degree.

Specifically, the debugging process revealed that a higher level of analytical and problem-solving skills is necessary to contribute fully to complex, real-world development projects. Ultimately, overcoming these coding challenges and successfully deploying a functional intrusion detection system has provided me with new insights and a massive surge of motivation to pursue a dedicated career in machine learning and network security.

CHAPTER 7

Conclusion and Future Work

7.1 Conclusion

In this work, I am focused on a very practical question: how can we effectively protect smart medical devices from cyberattacks? By specifically targeting ARP Spoofing within the Internet of Medical Things (IoMT), this project tackles a real-world problem that directly impacts patient data and safety. Through building and testing a complete machine learning pipeline, I learned that securing these networks does not necessarily require the most complicated or resource-heavy algorithms. After processing the CICIoMT2024 dataset[2] and evaluating seven different models, the Random Forest algorithm stood out as the clear winner, achieving an impressive accuracy of 96.2%.

It proved that tree-based models are incredibly reliable at filtering network traffic without causing too many false alarms. This balance is crucial in a hospital setting where doctors need constant, uninterrupted access to their medical devices.

Beyond the numbers, this project was a massive learning experience. Going from theoretical concepts to actually writing Python code, debugging cloud environment errors, and manually balancing datasets gave me genuine, hands-on confidence. It successfully bridged the gap between classroom learning and practical software engineering.

While the current results are highly encouraging, there is always room to grow and improve. The immediate next step as future work is to stress-test this machine learning framework on a massive scale by evaluating it against the entire dataset of millions of records, ensuring that the high accuracy holds up under heavy, enterprise-level loads. Beyond offline testing, the ultimate goal is to physically deploy this lightweight model directly into real-world hospitals. By running the system on cost-effective edge devices, such as a Raspberry Pi plugged into a medical router, we can actively detect and block live cyber threats right next to the patient monitors. These directions promise to advance both our understanding of IoMT systems and the practical capabilities of cybersecurity defenses in the medical field. In models on the CICIoMT2024 dataset[2], it was demonstrated that effective security does not

require resource-heavy algorithms. The Random Forest classifier emerged as the optimal solution, achieving an impressive 96.2% accuracy. It proved highly reliable at filtering network traffic with minimal false alarms—a crucial balance for maintaining uninterrupted access to life-saving medical devices.

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